MILITARY SPECIFICATION SHEET

# ELECTRON TUBE, GAS

## TYPE 83

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Rectifier, full-wave, mercury-vapor

Envelope: 16-1 (EIA)

Base: A4-9, phenolic (EIA)

Pin connections:

| Pin No. | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| Element | f | a2 | a1 | f |

Mounting position: Any

Weight: 5-ounces (141.75 grams) nominal

ABSOLUTE RATINGS:

| Parameters: | Ef | epx | Epp | ib | Io | Rℓ | Cℓ | i surge | THg |
|---|---|---|---|---|---|---|---|---|---|
| Unit: | Vac | v | Vac | ma | mAdc | Ohms | µF | a | °C |
| Maximum: | 5.5 | 1,650 | --- | 800 | 250 | --- | --- | 2.50 (see note 1) | +70 |
| Minimum: | 4.5 | --- | --- | --- | --- | --- | --- | --- | +20 |
| TEST CONDITIONS: | 5.0 | --- | 500 | --- | --- | 2,500 | 8 | --- | --- |

GENERAL:

Qualification - Required

© denotes changes



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| METHOD | REQUIREMENT OR TEST | NOTES | CONDITIONS | AQL (PERCENT DEFECTIVE) | INSPECTION LEVEL OR CODE | SYMBOL | LIMITS MIN | LIMITS MAX | UNIT |
|---|---|---|---|---|---|---|---|---|---|
| | Quality conformance inspection, part 1 | | | | | | | | |
| --- | Dimensions | - | Base: A4-9 phenolic | 1.0 | II | --- | --- | --- | --- |
| 1353 | Operation of rectifiers | 2,4 | | 0.65 | II | Io | 235 | --- | mAdc |
| 1346 | Voltage drop | - | Ebb = 125 Vdc; Rt/I1b+I2b = 1,000 mAdc; THg = 20° to 60°C | 0.65<br>0.65 | II<br>II | E1b<br>E2b | ---<br>--- | 18<br>18 | Vdc<br>Vdc |
| 1301 | Filament current | - | | 0.65 | II | If | 2.7 | 3.3 | Aac |
| | Quality conformance inspection, part 3 | | | | | | | | |
| --- | Life test | 3 | Group B; t = 500 hours | --- | --- | --- | --- | --- | --- |
| --- | Life-test end point: | | | | | | | | |
| 1353 | Operation of rectifiers | - | | --- | --- | Io | 210 | --- | mAdc |

NOTES:

1. Ratings apply to a power-supply frequency below 150 Hz and to an ambient temperature of condensed mercury vapor between 20° and 60°C.
2. In a full-wave circuit adjust Zp/p such that a tube having Etd = 15 Vdc at 800 mAdc per anode gives Io = 250 mAdc.
3. In life-test operation, the values of Rt and Ct given in test conditions may be considered as approximate, and shall be adjusted initially to give Io = 250 mAdc with ib equal to or greater than 720 ma.
4. This test shall be the first test performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2925)



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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

**DD FORM 1 JAN 72 1426** REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED S/N 0102-014-

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